AP PHYSICS C

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0.1 Foreword for anybody else using this

As of writing this, I've already taken AP Physics 1 and 2 and Calc BC, so I'll skim over those stuff. I steal the images from google. If this violates copyright somehow, please email me and I'll remove it.

1 Unit 1: Kinematics

1.1 Scalars and Vectors

Scalars are numbers. Vectors are vectors. Represent as $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$.

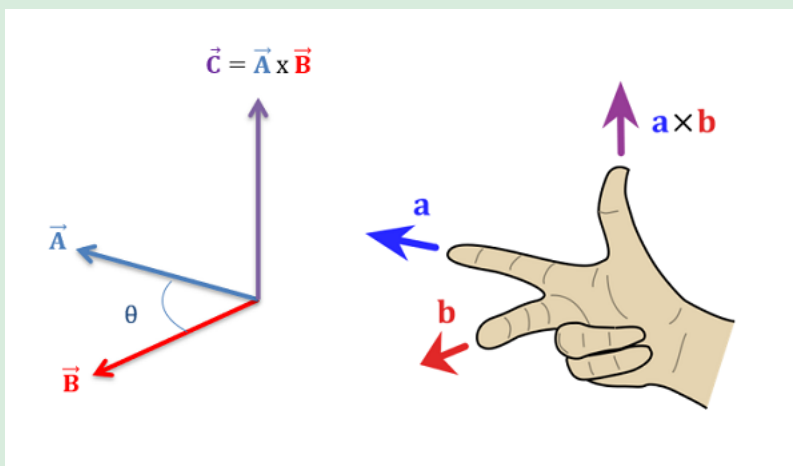
Dot product of two vectors is the length of one of the vectors projected onto the other multiplied by the length of said other vector. Theta = min angle between two vectors.

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

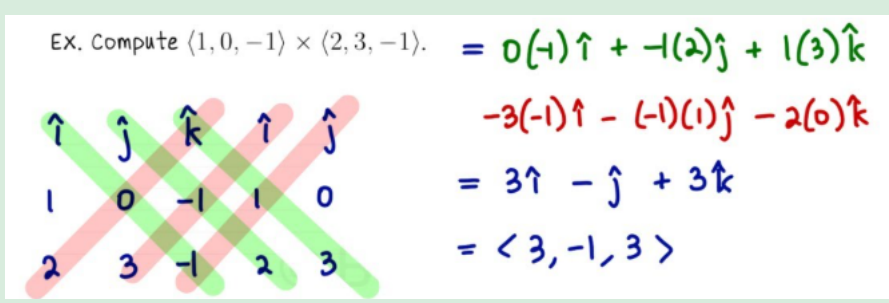
1.1.1 Cross Product

Operation that can be applied to two **three-dimensional** vectors to generate a third vector perpendicular to both vectors. Use the **right hand rule** to deterimine which direction the third vector will go.

Principle 1.1 (Right Hand Rule)
To find the geometrical direction of $\vec{a} \times \vec{b}$, hold your right hand out and point your *index* finger in the direction of $\vec{a}$, and your *middle* in the direction of $\vec{b}$. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



Principle 1.2 (Calculating the Cross Product)
To calculate the cross product between $\vec{A}$ and $\vec{B}$:



Note the following:

1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ where theta is the minimum angle betwen the two vectors. Also equal to the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
2. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. This property is known as the **anti-commutativity of the cross product**.
3. $\vec{a} \times \vec{a} = \vec{0}$ $(\theta = 0)$

1.2 Displacement, Velocity, and Acceleration

$$\vec{v} = \frac{\mathrm{d}\vec{s}}{\mathrm{d}t}, a = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{s}}{\mathrm{d}t^2} \implies \Delta\vec{s} = \int \vec{v}(t)dt, \Delta\vec{v} = \int \vec{a}(t)dt$$

Derivatives and intergrals of vector-valued functions can be done component wise.

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \frac{\mathrm{d}\vec{s}}{\mathrm{d}t} = x'(t)\hat{\imath} + y'(t)\hat{\jmath} + z'(t)\hat{k}$$

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \int_{t_i}^{t_f} \vec{s}dt = \int_{t_i}^{t_f} x(t)dt + \int_{t_i}^{t_f} y(t)dt + \int_{t_i}^{t_f} z(t)dt$$

For constant one-dimensional acceleration, kinematic equations apply:

$$\Delta v = \int adt \implies v_f = v_i + at$$

$$\Delta s = \int v(t)dt = \int (v_i + at)dt \implies \Delta s = v_i t + \frac{1}{2}at^2$$

$$v_f = v_i + at \implies v_f^2 = v_i^2 + 2v_i at + a^2t^2 \implies v_f^2 = v_i^2 + a(2v_i t + at^2) \implies v_f^2 = v_i^2 + 2a\Delta x$$

Falling object experience constant acceleration: $a = g \approx 9.81\mathrm{m\,s^{-2}} \approx 10\mathrm{m\,s^{-2}}$. Projectiles experience constant vertical accleration of gravity, but no horizontal acceleration.

1.3 Representing Motion

Distance, velocity, and acceleration vs. time graphs exist. Yipee!

1.4 Reference Frames and Relative Motion

Let $\vec{v}_{AB}$ = velocity of A relative to B.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

These are also known as **Galilean Transformations.** Note these equations will break down near the speed of light due to relativity shit.

Example 1.3
The president's airplane flies at $250\mathrm{m\,s^{-1}}$ to the east with respect to the air. The air is moving at $35\mathrm{m\,s^{-1}}$ to the north with respect to the ground. Find the velocity of the president's airplane with respect to the ground.

Answer 1

Let $\hat{\imath}$ = north and $\hat{\jmath}$ = east.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} = (250\hat{\jmath} + 35\hat{\imath})$$

$$|\vec{v}_{PG}| = \sqrt{250^2 + 35^2} \approx 252.4\,\mathrm{m\,s^{-1}}$$

$$\theta_{PG} = \arctan\frac{35}{250} \approx 7.97^\circ \text{ relative to East}$$

1.5 Motion in Two or Three Dimensions

everything that we've talked about already considers three dimensions :3

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8.1 Electric Charge and the Electric Force

Charge... exists. It's scalar and measured in SI units **Coulombs**.

Coulomb's Law describes the electrostatic force between two charged particles:

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

ϵ_0 represents the **electric permittivity of free space** and has a constant value (units $C^2N^{-1}m^{-2}$). Electric permittivity measures "the degree to which a material or medium is polarized in the presence of an electric field" and is determined by "the ease with which electrons can change configurations within a given material." Electric polarization can be modeled as the induced rearangement of electrons by an external electric field.

8.2 Conservation of Electric Charge and the Process of Charging

8.3 Electric Fields

8.4 Electric Fields of Charge Distributions

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